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ARI5118 - Deep Learning for Computer Vision Assignment

Feature Pyramid Networks and Multi-Scale Detection

By

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1.0 Introduction

Object detection is a quintessential problem in computer vision and, arguably, the most explored of all. Some of the many applications are self-driving cars, medical image diagnostics, analysis of aerial and satellite images, and inspection of industrial products. At its heart, detection needs a system that simultaneously answers two questions for each position in the image: is there an object here? What object is this? Although the advent of deep convolutional neural network architectures has revolutionized the field, achieving accuracy well above the classical methods in general, one problem seems surprisingly stubborn: the scale issue.

1.2 The Scale Problem Defined

The scale problem is a consequence of the inherent compromise between deep convolutional networks and scale. As an image passes through a CNN's layers of convolution and pooling, two things change in opposing ways. Firstly, spatial resolution diminishes; earlier layers are sensitive to every pixel in an image; the final layers that are fed to the classifier each have an enormously large receptive field, so they cannot capture detail about tiny spatial structures. Secondly, the semantic level increases: later layers learn enormously rich, high-level, semantic information, which is valuable for classification, but there are relatively poor, spatial representations available to enable small object localization.

This produces an intrinsic accuracy gap. Earlier features contain the localization detail required to find small objects, but do not provide the semantic features needed for classification. Later layers, on the other hand, do have sufficient semantic representation for classification but due to the large amount of pooling applied to them, have very low spatial resolution. A straightforward single-scale detector which just uses the final feature map gives great results on large objects but very poor or non-existent performance on small and medium objects. This is a real issue: on COCO (Common Objects in Context) (Lin et al., 2014), performance is evaluated separately for small (area < 32 pixels), medium and large objects, and there are substantial gaps between state-of-the-art performance on different sizes.

1.3 Pre – FPN Approaches and Their Limitations

Prior to the use of Feature Pyramid Networks (FPNs), two categories of approaches existed to solve the scale problem. First, featurized image pyramids treated the problem of different scales by running the detector on several re-scaled versions of the input image. However, this is computationally expensive: since the computational cost doubles for each doubling of scale, the complexity quadruples for each doubling, which is not viable in real-time and for large scale applications.

The second category, multi-scale feature hierarchies, exploit the fact that CNNs, by nature of down-sampling, naturally extract feature maps at various resolutions. Single Stage Detector (Liu et al., 2016) and Multi Scale-CNN (Cai et al., 2016) take advantage of this hierarchy by detecting on multi-level features, appending detection heads to several intermediary feature maps. In doing so, they discard the information in neighbouring layers that would be complementary; instead, each scale operates in relative isolation from the others. The shallowest layers of the highest resolution were not even considered, as their weak semantics

caused more harm than good, and as is common in many proposals; single-stage object detector avoided this feature map as well.

1.4 Feature Pyramid Network as a Principled Solution

Introduced by Lin et al. (2017), feature pyramids have been proven powerful in image understanding tasks that require multi-scale features. State-of-the-art methods for multi-scale feature learning focus on performing feature interactions across space and scales using neural networks with a fixed topology. Feature pyramids display high-level feature maps of various scales and collaborate with backbone networks to attain more balanced and enhanced performance in both object detection and semantic segmentation across multiple scales. Feature pyramids have been recently developed in three main forms: top-down networks, top-down/bottom-up networks and attention-based methods. Feature pyramid network (FPN) leverages the natural multi-scale, pyramidal structure of deep convolutional neural networks, and creates a top-down architecture with lateral connections to gain high-level semantic feature maps in all scales.

Since its inception, FPNs have seen continued development. The original framework has been extensively enhanced, modified, and evaluated by many authors since its release. A bottom-up path to augment features was subsequently incorporated by PANet (Liu et al., 2018). Graph FPN (Zhao et al., 2021) abandons the fixed grid framework and, instead, takes a graph structure based on the image topology, such that feature interactions occur concurrently at all scales rather than adjacent scales. Feature aggregation is instead pushed into a latent graph-convolutional space by LFPN (Xie et al., 2022) so that long-range features can be represented, an aspect that can't be achieved using point-wise convolution. Triangular cross-layer aggregations are proposed in MSRA-FPN (Wang and Wang, 2023) to compensate for information loss incurred when scaling the features up, and domain specific FPN variations such as DM-FPN (Zhang et al., 2019) attempt to tackle issues that are present when processing very-high resolution data in remote sensing images.

1.5 Assignment Scope

This assignment provides a comprehensive treatment of Feature Pyramid Networks and multi-scale detection from first principles to real-world application. Section 2 offers a critical synthesis of the key literature, tracing the conceptual evolution from the original FPN to its modern successors. Section 3 presents the technical foundations in depth, covering the architecture's three structural components, the underlying mathematics, a fully worked numerical example at each pyramid level, and an analysis of common misconceptions. Section 4 connects these technical concepts to concrete applications, examining how FPNs enable high-performance detection in geospatial imaging, small-object scenarios, medical analysis, and beyond. The assignment concludes with a curated bibliography that includes the primary papers examined throughout.

2.0 Literature Review

One of the most fruitful areas of computer vision research over the last ten years has been the creation of multi-scale feature representations in deep learning. The field's development from the initial conception to graph-based, latent-space, residual-aggregation, and remote-sensing-specific extensions is traced in this section, which summarizes five foundational and extension studies pertinent to FPNs and multi-scale detection. Starting with the architectural base and moving on to progressively specialized elaborations, the articles are analyzed thematically rather than chronologically.

2.1 Feature Pyramid Networks for Object Detection, Lin et al. (2017): The Founding Architecture

The seminal paper by Lin, Dollar, Girshick, He, Hariharan, and Belongie, "Feature Pyramid Networks for Object Detection" (CVPR 2017) provided the theoretical framework and practical implementation over which all subsequent efforts have been built. The key insight articulated was that although the inherent nature of CNNs is to produce features at multiple resolutions, neither single-scale detectors nor methods akin to SSD simultaneously achieve the convergence of strong semantic content and resolution at each of the multi-scale feature maps.

The approach that they introduced of using top-down pathway with lateral connections was surprisingly straightforward. Through a sequence of down-sampling and fusion, the deepest network features are diffused downwards, such that at each level in the output pyramid we have information of the most semantics (deepest layer of backbone) with the high spatial resolution of the shallowest layers. FPN based Faster R-CNN has achieved the best among all single-model entries published at the time of publication with a COCO dataset. The authors improved the Average Precision by 2.3 points over a solid baseline of ResNet with almost no additional inference cost. Furthermore, they showed their FPN is backbone-independent, and also a strong proposal network in instance segmentation. It has revolutionized the field, and currently FPN can be considered as an architecture to be included in most of the state-of-the-art object detectors.

For this assignment, the Lin et al. Paper provides the primary technical reference and all the elements described in section 3 are based on their definition. The ablation study results presented in the paper-examining the independent effect of the top-down pathways, lateral connections, and pyramid representation-serve as the evidential basis for the assertions made herein.

2.2 Graph-Based Feature Interaction, Zhao et al. (2021)

Graph Feature Pyramid Network for Object Detection' (ICCV 2021) by Zhao, Ge, and Yu is a fundamental rethinking of the interaction between multi-scale features. Conventional FPNs are primarily criticized by the authors for forcing a predetermined grid topology on a process that ought to adjust to the inherent content of the image. Objects in real photographs are not evenly distributed throughout a spatial grid; instead, they cluster, overlap, and exist in intricate part-whole connections that are impossible to represent with a static topology.

GraphFPN solves this problem by creating a superpixel hierarchy for each input image using hierarchical segmentation (COB, Maninis et al., 2017) and then constructing a graph pyramid where nodes at different levels in the pyramid correspond to superpixels at different resolutions. We define two types of graph edges: contextual edges between adjacent nodes at the same resolution, and hierarchical edges between nodes at different resolutions that are ancestors or descendants of each other. Graph neural network with self-attention and local channel attention runs on top of this variable-topology graph structure and allows features to interact globally at each level instead of just between adjacent levels. It achieves an AP of 43.7 on MS-COCO 2017 test-dev, outperforming FPN-based methods and recent transformer-based detectors. It yields particular good results on small objects AP.

This work is important to us for three reasons. First, it proves that the fixed-topology assumption of traditional FPNs is actually a limitation instead of just a design choice. Second, the adoption of superpixel hierarchies to capture image-specific structure foreshadows key concepts that become essential to vision transformers later on. Third, its ablation experiments provide evidence that both within-level contextual information and across-level hierarchical interaction are crucial to the multi-level detection tasks, when removing either contextual layer or hierarchical layer, the performance drops dramatically.

2.3 Latent Feature Pyramid Network for Object Detection, Xie et al. (2022)

Xie, Pang, Nie, Cao and Han's "Latent Feature Pyramid Network for Object Detection" (IEEE Transactions on Multimedia, 2022) noted that while standard FPNs have reduced the semantic gap by introducing top-down aggregation and cross-scale connections, they still employ only element-wise fusion strategies like addition, concatenation and weighted multiplication. While efficient, such operations do not model long-range spatial relationships; they can only consider each feature vector in isolation from others in its own feature map, no matter how spatially distant the associated pixels. The two pixels corresponding to faraway, semantically similar feature vectors will only contribute to fusion individually. The LFPN attempts to correct this limitation. It first projects feature maps from the spatial domain into a low-dimensional latent space by training a projection matrix, then applies a GCN to fuse the feature maps in latent space, and finally reverse-projects them back to the spatial domain. The method proposes two types of fusion: an Inner-Scale Fusion (ISF) to model long-range dependencies in a single pyramid level, and a Cross-Scale Fusion (CSF) to jointly model dependencies among pyramid levels. Operating in latent space makes GCN computationally feasible because the latent space dimension is drastically smaller than the number of pixels in any given feature map. LFPN is implemented as a plug-in module and works with any existing FPN and any backbones; it is tested on MS-COCO and results in about 1% AP improvement with little inference overhead. The LFPN also relates to the discussion in Section 3 in the way it acknowledges explicitly what element-wise operations take for granted, namely the dependencies between features at different pyramid levels. Furthermore, the use of latent space connects FPN with a field of graph based reasoning in deep learning.

2.4 Multi-Scale Residual Aggregation Feature Pyramid Network for Object Detection, Wang and Wang, (2023)

In 'Multi-Scale Residual Aggregation Feature Pyramid Network for Object Detection' (Wang and Wang, 2023), they highlight a particularly vexatious weakness of standard FPNs, the degradation of information in the upsampling stage. When deep layer features are upsampled and summed into shallow layer features using standard nearest-neighbor interpolation then element-wise addition, these deep features gradually weaken as they traverse up multiple stages, reducing the semantic quality of the lower-level pyramid features, thereby affecting especially large object detection negatively.

The MSRA-FPN offers three means to overcome these weaknesses, a Uidirectional Cross-Level Residual Aggregation Module (UCLRAM), where the triangular bottom-up aggregation of features is influenced by deep layer aggregation (DLA) to combine the raw features at various scales within the backbone at the uppermost layer prior to the top-down processing; the top-down path includes residual connections to maintain high-level features at each level of the pyramid, and a Residual Squeeze and Excitation Module (RSEM), where channel attention is utilized at the top-down fusion nodes to diminish alias effects due to merging features at the various scales; overall, MSRA-FPN yielded improvements of between 0.5 and 1.9 percentage points over the baseline detector on PASCAL VOC and MS-COCO, as well as substantial improvement in the detection of large objects and on a custom Thangka Figure Dataset.

The most relevant aspect of this work with respect to the literature review is the explicit emphasis on the imbalance in the semantic quality of features at different pyramid levels; this issue is also present in other works, e.g. AugFPN (Guo et al., 2020), but in a more structured and formalized manner here. The triangular aggregation structure employed here is also innovative in relation to standard FPN top-down pathways as well as PANet bottom-up pathway extensions.

2.6 Geospatial Object Detection on High Resolution Remote Sensing Imagery Based on Double Multi-Scale Feature Pyramid Network, Zhang et al. (2019)

The "Geospatial Object Detection on High Resolution Remote Sensing Imagery Based on Double Multi-Scale Feature Pyramid Network" by Zhang, Zhu, Chen, Tan, Zhang, Dai, Liao and Gong, Remote Sensing, 2019, represents a good example of how the FPN framework needs to be adapted in circumstances where the task imposes limitations quite different from natural object detection. When dealing with very-high-resolution (VHR) imagery, there are many tasks in combination where the issues are much harder than for standard benchmarks: images can be of the size thousands of pixels across, only top-down views of objects are available, where there are no canonical orientations, object distributions are heavily imbalanced, in the order of hundreds of small objects versus one large building, and ground truth is defined by oriented bounding boxes rather than axis-aligned rectangles.

DM-FPN deals with these issues by using FPN both in the region proposal network and in the class-specific detection network of a Faster R-CNN framework. RoI Align, proposed in [11],

is used instead of RoI Pooling to get rid of the quantization errors introduced by the former. These quantization errors have a more negative effect on small object detection than on larger objects and have a huge impact in this scenario. Furthermore, the authors suggest using patch-based multi-scale training and inference: this means, in this case, to use overlapped tiles (with some overlapping region) at multiple scales for each large image to increase data diversity. The presented technique of Adaptive Categorical Non-Maximum Suppression (ACNMS) works by adapting the intersection over union (IoU) suppression threshold according to object density for each category, which avoids the over-suppression of actually overlapped dense objects, such as vehicles in a parking lot. The DM-FPN reaches a state-of-the-art mAP score of 0.793 in the DOTA aerial object detection benchmark, which is superior to all other state-of-the-art works reported in this work.

This paper is used not to highlight an architectural innovation per se but more to show how FPN adapts to harsh deployment conditions, and its results, detailed further in Section 4, show that multi-scale features are a must when dealing with real high-stakes applications, not only in academia.

2.7 Synthesis and Research Agenda

Overall, the 5 papers present a clear flow of research. While Lin et al. (2017) provides a general framework, other studies attempt to resolve different bottlenecks: GraphFPN does not assume a fixed topology; LFPN deals with the gap of long-range dependence induced by element-wise fusion; MSRA-FPN combats the information degeneration along the upsampling path; DM-FPN conducts domain adaptation in remote sensing imagery. The common message delivered by these 5 papers is that single-scale detection per se is inefficient and that the mechanism of fusing multi-scale information is crucial, rather than simply having multi-scale features, which prompts the technical description below.

3.0 Technical Foundations and Core Algorithms

The Feature Pyramid Network (FPN) architecture serves as the "neck" of modern object detectors, bridging the gap between high-level semantic knowledge and low-level spatial precision. It transforms the standard backbone of a Convolutional Neural Network (CNN) into a multi-scale feature extractor where all levels are semantically strong.

3.1 FPN Architecture Overview

There are 3 structural parts within FPN: bottom up pathways (backbone CNN), top down pathway (a sequence of upsampling and merge operations) and lateral connections (a set of 1x1 convolutions to project features from backbone to same channel dimension, then merge). All three parts are merged to output a feature pyramid $\{P_2, P_3, P_4, P_5\}$. The output levels P_2, P_3, P_4, P_5 , have the same resolution of backbone features, but with richer semantics. We use the analogy as: Backbone acts like a reader skimming the image, sacrificing detail to extract meaning from the high-level context and lose much of local fine-grained details. The top-down pathways act like a second read over the image, now that a general semantic understanding has been obtained from the first pass, a second pass over the image can allow for each region to be classified using higher-level semantic information.

- **Bottom-Up Pathway:** This is the feed-forward computation of the backbone (e.g., ResNet). As the image moves deeper, it is down-sampled, increasing the receptive field and semantic richness while losing spatial detail. The output of the last residual block in each stage is denoted as $\{P_2, P_3, P_4, P_5\}$ representing strides of $\{4, 8, 16, 32\}$ pixels respectively.
- **Top-Down Pathway:** This pathway starts from the semantically rich P_5 and hallucinates higher-resolution features by upsampling coarser maps by a factor of two. This propagates high-level context back to larger spatial resolutions.
- **Lateral Connections:** To recover the precise localisation lost during down-sampling, 1×1 convolutions are applied to the bottom-up maps to reduce their channel dimensions to match the top-down maps. These are then merged via element-wise addition.

3.2 Mathematical Foundations of Multi-Scale Feature Extraction

To see why FPN is effective, consider how convolutional layers operate differently across pyramid levels. A single convolutional layer with kernel size k applied to a feature map of spatial resolution $s \times s$ produces an output of spatial resolution approximately $s \times s$ (with appropriate padding). Its receptive field — the region of the original input image that contributes to a single output value — grows with each successive layer.

After L layers with strides $s_1, s_2, \dots, s_L$, the effective receptive field has a footprint proportional to the product of all strides. For ResNet-50, this means that a single value in C5 integrates information from a 32×32 pixel region of the original image. A small object occupying 8×8 pixels is represented by approximately $(8/32)^2 \approx 6\%$ of a single C5 cell —

effectively invisible. The same object spans $(8/4)^2 = 4$ cells in C2, providing sufficient spatial resolution for localisation.

The lateral connection in FPN can be understood as a form of residual addition in the spatial hierarchy. The top-down upsampled features T_{i+1} carry semantic context from the entire image. The lateral projection $L_i = W_i * C_i$ (where W_i is the 1×1 weight matrix) selects a semantically useful subspace of the backbone features. Their element-wise sum $P_i = T_{i+1} + L_i$ thus integrates two complementary information streams: global semantic context (from T) and local spatial precision (from L). The subsequent 3×3 convolution fuses these two streams into a unified representation.

In formal terms, FPN can be viewed as a multi-scale encoder-decoder. The backbone is the encoder, which progressively increases semantic content while reducing resolution. The top-down pathway is the decoder, which progressively restores resolution while preserving semantic content. Unlike typical encoder-decoders for segmentation (such as U-Net), FPN produces predictions at every decoder level rather than only at the finest resolution. This makes it better suited to detection, where objects of different sizes require different resolution-semantic tradeoffs.

3.3 Worked Numerical Example: FPN Feature Propagation

The following worked example traces the construction of a two-level FPN pyramid (P3 and P4 only, for clarity) through a simplified backbone. Dimensions are chosen to be computationally tractable while preserving the essential structure of the real architecture.

Setup

Assume an input image of $64 \times 64 \times 3$. The backbone produces:

- C3: $8 \times 8 \times 512$ (stride 8 — fine resolution, weaker semantics)
- C4: $4 \times 4 \times 1024$ (stride 16 — coarser resolution, stronger semantics)

Target pyramid channel dimension: $d = 256$ channels.

Step 1: Build P5 (top-most level)

A 1×1 convolution projects C4 to 256 channels, creating the initial top-down signal:

$$T4 = \text{Conv}_{1 \times 1}(C4) \quad [4 \times 4 \times 1024 \rightarrow 4 \times 4 \times 256]$$

P4 is formed by applying a 3×3 convolution to T4:

$$P4 = \text{Conv}_{3 \times 3}(T4) \quad [4 \times 4 \times 256 \rightarrow 4 \times 4 \times 256]$$

Step 2: Propagate to P3

The top-down feature T4 is upsampled by a factor of 2:

$$\text{Upsample}(T4) \quad [4 \times 4 \times 256 \rightarrow 8 \times 8 \times 256]$$

The backbone feature C3 is projected to 256 channels via a separate 1×1 convolution:

$$L3 = \text{Conv}1 \times 1(C3) \quad [8 \times 8 \times 512 \rightarrow 8 \times 8 \times 256]$$

The lateral connection merges these two signals by element-wise addition:

$$\text{Merged3} = \text{Upsample}(T4) + L3 \quad [8 \times 8 \times 256]$$

A 3×3 convolution smooths the merged features:

$$P3 = \text{Conv}3 \times 3(\text{Merged3}) \quad [8 \times 8 \times 256]$$

Step 3: Numerical illustration of a single channel

To make the above concrete, consider a single channel of T4 after upsampling and a single channel of L3 at a 4×4 patch within the 8×8 merged feature map. Suppose the values at a 2×2 block are:

$$\text{Upsample}(T4)[0:2, 0:2] = [[0.8, 0.8], [0.8, 0.8]] \quad (\text{nearest-neighbour duplication})$$

$$L3[0:2, 0:2] = [[0.3, 0.5], [0.2, 0.6]] \quad (\text{lateral projection from C3})$$

$$\text{Merged3}[0:2, 0:2] = [[1.1, 1.3], [1.0, 1.4]] \quad (\text{element-wise addition})$$

The merged values now carry both the abstract semantic signal from the deep feature T4 (uniformly 0.8, reflecting high-level context) and the precise spatial signal from the backbone C3 (varying by position, reflecting local structure). A subsequent 3×3 convolution integrates these values across their spatial neighbourhood, producing the final P3 representation that combines both properties.

Step 4: Anchor Assignment

For a proposed region of interest with width $w = 12$ pixels and height $h = 10$ pixels in the original 64×64 image:

$$\begin{aligned} k &= \text{floor}(4 + \log_2(\sqrt{12 * 10} / 224)) \\ &= \text{floor}(4 + \log_2(10.95 / 224)) \\ &= \text{floor}(4 + \log_2(0.0489)) \\ &= \text{floor}(4 + (-4.35)) \\ &= \text{floor}(-0.35) = -1 \rightarrow \text{clipped to P2} \end{aligned}$$

This small object (approximately 11×10 pixels) is assigned to the finest pyramid level P2 (or P3 if P2 is excluded). A large object with $w = 200$ and $h = 180$ would compute $k = \text{floor}(4 + \log_2(190/224)) = \text{floor}(4 + (-0.24)) = 3$, assigning it to P3, which has coarser resolution but stronger semantics — appropriate for its size.

Step 5: Parameter count

The lateral 1×1 convolutions contribute $512 \times 256 + 1024 \times 256 = 131,072 + 262,144 = 393,216$ parameters in this two-level example. The two 3×3 post-merge convolutions contribute $256 \times 256 \times 9 \times 2 = 1,179,648$ parameters. These are small relative to the backbone (ResNet-50

has approximately 25 million parameters), confirming the FPN's characteristic of adding significant representational power at modest computational cost.

4.0 Real-World Applications in Computer Vision

These architectural advantages of FPNs: rich multi-scale feature information, computational efficiency and detector independence - manifest into direct performance benefits in several applied computer vision fields. Section 4 is concerned with four such application areas in which multi-scale detection in FPNs is highly significant and relates the mechanics described in Section 3 to the application challenges in each case.

4.1 Geospatial and Aerial Object Detection

The multi-scale object detection application that has proved to be among the most challenging is that of remote sensing and aerial imagery analysis. This is evidenced by the DOTA benchmark (Xia et al. 2018, which Zhang et al. 2019 utilized as the main evaluation framework) comprising 188,282 annotations of objects across 15 categories ranging from objects on the scale of ground track fields (hundreds of meters in the aerial view) up to individual vehicles or helicopters. Aerial images can be simultaneously cluttered with instances of both types of objects and their size ratios can vary from 10,000:1 and higher in pixel footprint.

The top-down mechanism presented in FPN (see Section 3.1) is the very system that allows DM-FPN to excel on this task. By incorporating information from the higher layers back into the coarser feature maps, the fine-resolution feature map utilized in detecting the tiny vehicles (seen at P2/P3 level) still possesses semantics learned at the higher layers about the entirety of the image. Otherwise, the low-level features that successfully localize the vehicles would misclassify them due to surrounding clutter.

The application of FPN based models is not limited to DOTA alone. Operational remote sensing systems for the tasks of maritime traffic monitoring, damage detection in agriculture, urban sprawl detection or rapid damage assessment in disaster situations also extensively rely on these models. It must be emphasized that invariance to the object scale provided by the pyramid structure in this scenario is not merely a benefit but a necessity-an aerial imagery analysis tool that successfully identifies flooded areas as well as buildings in the area but failed to detect vehicles is operationally unfeasible.

4.2 Small Object Detection in Dense Scenes

The detection of small objects in crowded and messy environments represents the toughest task for a multi-scale detector. COCO defines 'small objects' as those smaller than 32 pixels in area for a reason: historically they are far slower than medium and large objects at detection. FPN had the largest gain (12.9 points) over a single-stage RPNbaseline on the COCO minival dataset in AR for small objects. It had a 12.9% AR improvement over the benchmark.

The reasoning, as mentioned in Section 3.1, is not hard to see: small objects are assigned to fine level pyramids (P2 or P3) where the resolution is highest and where features are less

smearred due to pooling. But lateral connections mix the features of the coarse levels (including P5 which holds strong semantic features), resulting in features with enough semantic information to classify even 12x8 objects. With coarse feature-enriched fine-level features, small-objects can be classified accurately instead of just localizing precisely.

For security and surveillance applications this is also extremely important. Facial recognition for monitoring security must find faces that are 20x20 pixels in wide field-of-view imagery. For detecting intrusion into a secured area, even small animals (and people) must be identified before entering the area even at a large distance. For both types of applications, small objects must be detected by FPN's enriched fine-level features.

To enhance this, GraphFPN (Zhao et al., 2021) introduced a super-pixel hierarchy to avoid feature contamination between foreground objects and background clutter, where the hierarchical nature effectively captures intrinsic image object boundaries which cannot be done on the current grid-based FPN. The improved APS (small object AP) achieved by GraphFPN over FPT is 2.3% which supports this structure.

4.3 Instance Segmentation

Predicting a bounding box (object detection) is only one downstream application supported by FPN-quality feature pyramids. Since the task of instance segmentation (predicting a pixel-wise mask over each object instance, as well as category and bounding box) places higher requirements on the feature representation than object detection, since a good mask needs both to know that something is a specific object (semantic content) and where each pixel belongs to the object (spatial information)-it can provide better quality representations.

Lin et al. (2017) validated FPN for instance segmentation by modifying the DeepMask/SharpMask network to use feature pyramids instead of image pyramids and obtained 48.1, an AR over instance 39.8 for SharpMask - a gap of over 8 points. Later, a Mask R-CNN (He et al., 2017) combined FPN with a mask prediction branch, which was later dominated instance segmentation and inspire many other downstream applications. A context of deployment where they are of particular importance is in medical imaging.

On a pathology slide, cellular structures at many scales can be of interest-cell nuclei, boundaries between cytoplasm and the medium, and overall organization of tissue at different scales; FPN-based segmenters have been applied to histopathology for tumor grading, polyp detection on colonoscopy video, and retinal vessel segmentation in fundus photography, all involve multi-scale structures where the initial idea for FPN is still highly relevant.

4.4 Autonomous Driving Perception

In autonomous driving applications, real time detection of objects that range over an incredibly vast scale variation within a single forward-facing camera view is needed: a traffic sign may be as small as 40 40 pixels while the vehicle is traveling at highway speed, whereas pedestrians may be as large as 100 300 pixels at some middle distance, and a truck as large as 400 200 pixels up close. There is limited computational budget available per frame due to real time

requirements (e.g., 30-60 Hz), and so architectures that are capable of covering this broad range of scales efficiently are highly desired.

FPN's architecture lends itself well to these requirements. The pyramid can be generated from previously computed internal representations in the backbone architecture, requiring no additional forward passes over the image, and thus adding only a marginal computation cost relative to a single-scale detection system. Lin et al. (2017) report 0.165s inference time per image for an FPN-based Faster R-CNN on a single GPU, comparable to the single-scale baseline of 0.32s, while significantly improving accuracy. In the setting of embedded-GPUs, such as those found in NVIDIA Drive platforms, optimized variants such as BiFPN (Tan et al., 2020), with their added weighted bi-directional connections and repetitive stacking of pyramid levels, also attain favorable accuracy-efficiency tradeoffs required for real time applications.

Latent FPN's claim, that it can effectively capture cross-scale, long-range dependencies between objects, is especially pertinent to autonomous driving. Objects such as a pedestrian within the proximity of a vehicle are co-dependent objects: the presence of one object implies the increased prior probability of the other object's existence. FPN inherently processes each object independently; LFPN's cross-scale fusion module explicitly accounts for these object interactions in a manner analogous to how human drivers would comprehend and predict their environments.

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Adversarial Quiz:

https://docs.google.com/forms/d/e/1FAIpQLSfqfDxvcKfdEJ7FCYpbgcmgTKMY1ogcnKg_BcHR0BOAvvopJQ/viewform?usp=sharing&ouid=102671720602482703339